COMPLETED COURSEWORK / PORTFOLIO REVISION

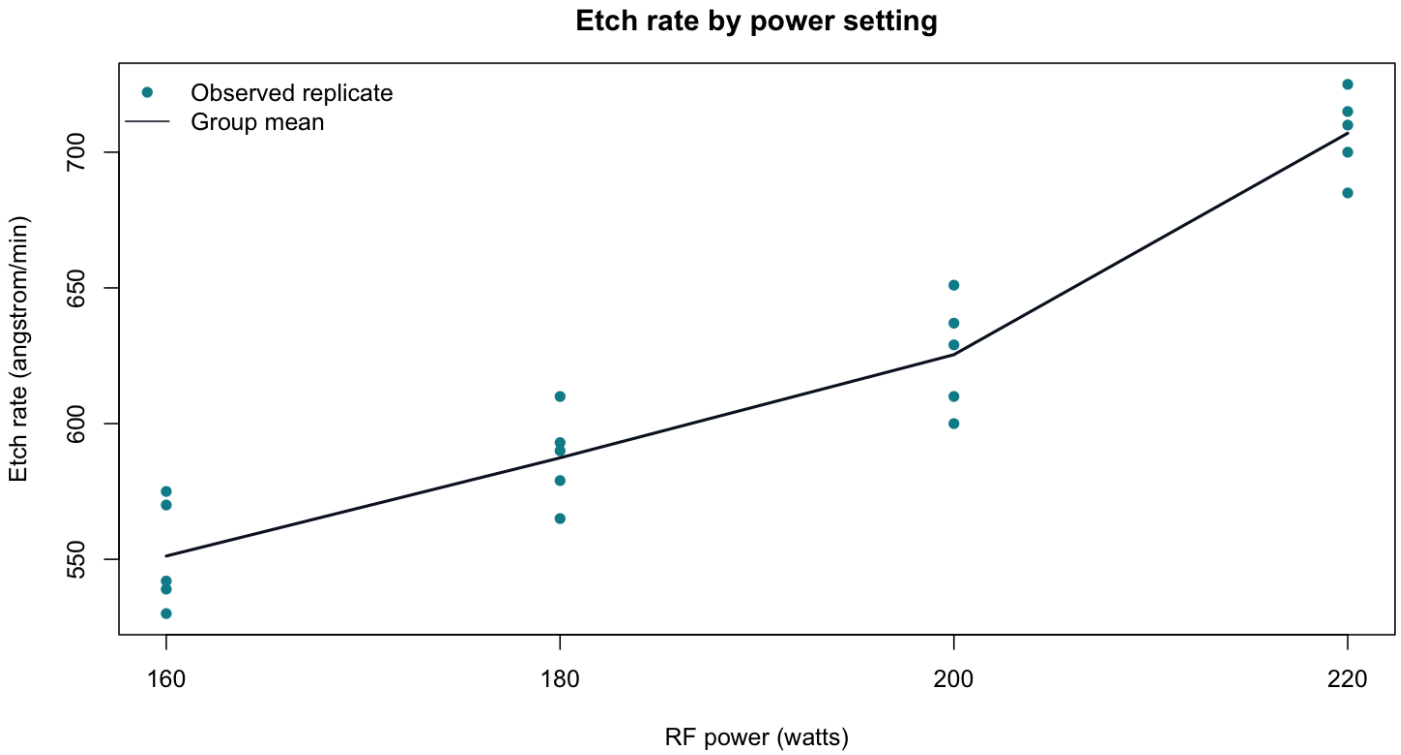
Experimental Design in R

RF power and plasma etch rate

Four power settings. Five replicates per setting.

JAMES OOKO

How does etch rate differ across RF power settings?



Mean etch rates rise from 551.2 at 160 W to 707.0 angstrom/min at 220 W. The latter is the highest tested mean, not a proven operating optimum.

Method, findings, and limits

Method

Treat power as a categorical factor. Fit a one-way ANOVA to 20 observations in four balanced groups. Inspect residuals and compare group means using Tukey-adjusted intervals.

Checked results

ANOVA: F(3,16) = 66.80, p = 2.88e-9. All six Tukey pairwise comparisons have adjusted p-values below 0.05. Shapiro-Wilk p = 0.215; Fligner-Killeen p = 0.791.

Interpretation limits

This is a textbook experiment reanalyzed for coursework, not an experiment I conducted. Independence and randomization depend on the source design. Small groups limit diagnostic power; a nonsignificant test does not prove assumptions. Results apply only to the tested settings and do not measure cost, uniformity, or wafer damage.

Source: course-provided plasma-etch numerical example from Montgomery, Design and Analysis of Experiments. Dataset values were checked against the local coursework; source documentation identifies four RF settings and five replicates.

Code and source notes: github.com/jamesooko/Final-Project-HOA732